# SRTF (Shortest Remaining Time first)

Criteria:- BT

· Tie-breaker: FCFS

· Type:- *Preemptive

Q:-

| Process | Arrival Time | Burst Time |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 4 |
| P3 | 2 | 1 |
| P4 | 3 | 2 |
| P5 | 4 | 1 |
| P6 | 5 | 3 |

| Process | BT |
|---|---|
| P1 | ~~6~~ 5 |
| P2 | ~~4~~ 3 |
| P3 | ~~1~~ 0 |
| P4 | ~~2~~ ~~1~~ 0 |
| P5 | ~~1~~ 0 |
| P6 | 3 |

Ready Queue:

| | |
|---|---|
| @ Time 0 | P1 |
| @ Time 1 | P1, P2 |
| @ Time 2 | P1, P2, P3 |
| @ Time 3 | P1, P2, P4 |
| @ Time 4 | P1, P2, P4, P5 |
| @ Time 5 | P1, P2, P6 |

| P1 | P2 | P3 | P4 | P5 | P2 | P6 | P1 |
|---|---|---|---|---|---|---|---|

0 1 2 3 ~~4~~ 5 6 9 12 17

| Process | AT | BT | CT | TAT | WT | RT |
|---|---|---|---|---|---|---|
| $P_1$ | 0 | 6 | 17 | 17 | 11 | 0 |
| $P_2$ | 1 | 4 | 9 | 8 | 4 | 0 |
| $P_3$ | 2 | 1 | 3 | 1 | 0 | 0 |
| $P_4$ | 3 | 2 | 5 | 2 | 0 | 0 |
| $P_5$ | 4 | 1 | 6 | 2 | 1 | 1 |
| $P_6$ | 5 | 3 | 12 | 7 | 4 | 4 |

Avg TAT = 37/6 ⇒ 6.16

Avg WT = 20/6 ⇒ 3.33

SL = 17 - 0 ⇒ 17

Th = 6/17

Response Time = ?

How many context switch ?

Q: Consider three CPU intensive processes, which require 10, 20, and 30 time units & arrival time 0, 2, 6 respectively. How many context-switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? (GATE-2006)

Do not count the context switches at time zero & at the end.

(A) 1 (B) 3

(C) 2 (D) 4

Q: An operating system uses Shortest remaining time first scheduling algo for pre-emptive scheduling of processes. Consider the following set of processes with their arrival times & CPU burst time (in ms) (GATE-2014)

| Process | Arrival time | Burst time |
|---|---|---|
| $P_1$ | 0 | 12 |
| $P_2$ | 2 | 4 |
| $P_3$ | 3 | 6 |
| $P_4$ | 8 | 5 |

Avg WT (in ms) = ?

⇒ Problems with SJF & SRTF:-

- Starvation
- No fairness
- Practical implementation is not possible